

I.T Exhibition – “Mili : The Robot”

It is a functional robot, that is capable of being controlled by a smartphone, of choice over bluetooth protocol. It is controlled using a application. It is controlled using Arduino boards, or “The brain of Mili is an Arduino”. It is capable of two dimensional or two axis movement, using its two motors, instead of four motors, for better motion capabilities.

It uses an “L298N Motor Driver”, which is controlled by an Arduino Nano (Fig. a) . Fig a shows the code that is used to control the Nano. We essentially are supplying voltage to both the motors in same direction for forward and backward control and supplying it either way for right and left control.

Figure (b) shows the schematic for the motor connection with the arduino, though here it shows a Arduino Uno, a Arduino Nano was used, but everything is the same regardless.

Locomotion – Mili’s steps! :-

As earlier said, it is made using two motors of torque of 1 Nm each, which totals to around 1 Nm or when converted to 2 kilogram, meaning that Mili can weigh under 2 kilograms. Both the motors are powered and driven using the L298N motor driver, which is controlled by the Arduino Nano.

Power – Mili’s Food! :-

It is powered by 3 x 9V batteries. They are in parallel connection, meaning the total voltage delivered is not 27 volts, rather 9 volts. The only change is in the power delivered, which is increases by 3 times. All the components, the Arduino Nano, The LED Matrix, The Bluetooth Chip as well as the motor driver and the motors are powered by the batteries.

Communication – Mili’s Nervous System! :-

An HC-05 Bluetooth Commute Module makes the communication between the phone and the arduino possible. A dedicated application has been made to make it possible (Fig c). Small string values are transmitted over the app to the chip, and the chip dumps the recieved values into the Digital Pins (2, 3), using the Software Serial Library. From there the values are read, and if appropriate values are supplied then the arduino commands the driver to move in the specified direction.

“f” – Forward Movement
“b” – Backward Movement
“l” – Left Movement
“r” – Right Movement
“s” – Pause Movement / Brake

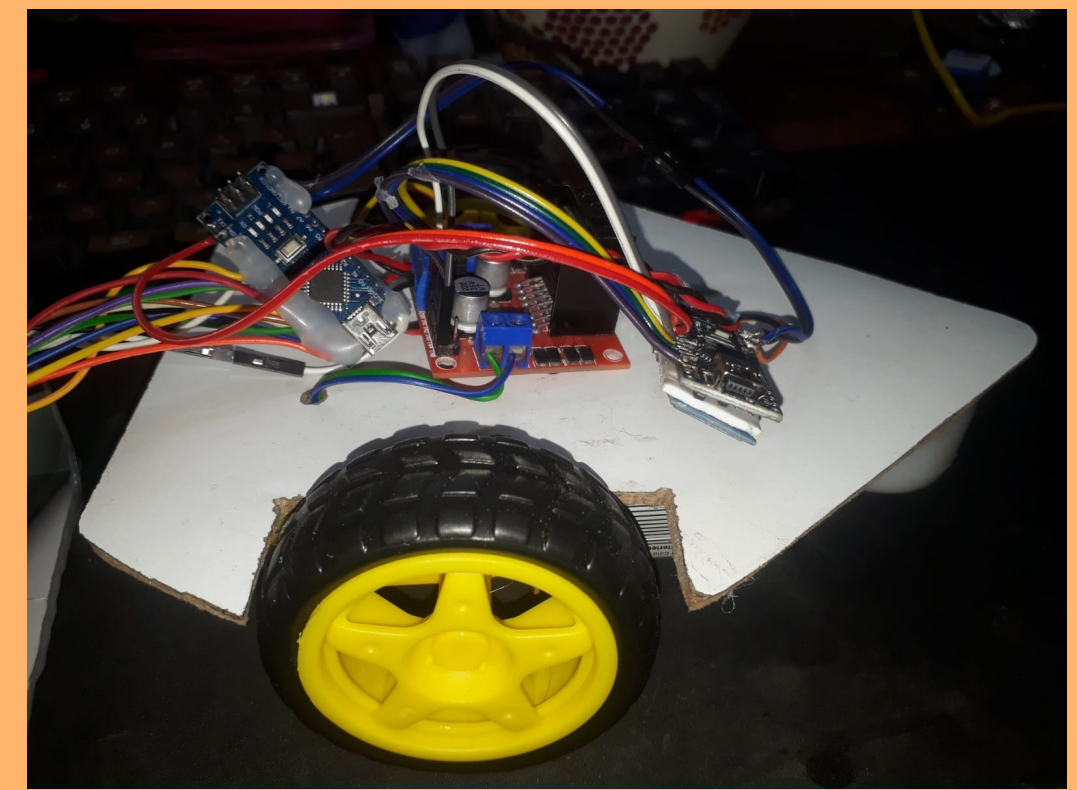
In the app, (fig d) , on clicking on the appropriate buttons, the respective values are sent over bluetooth to the Arduinno Nano.

LED Matrix – Mili’s Face! :-

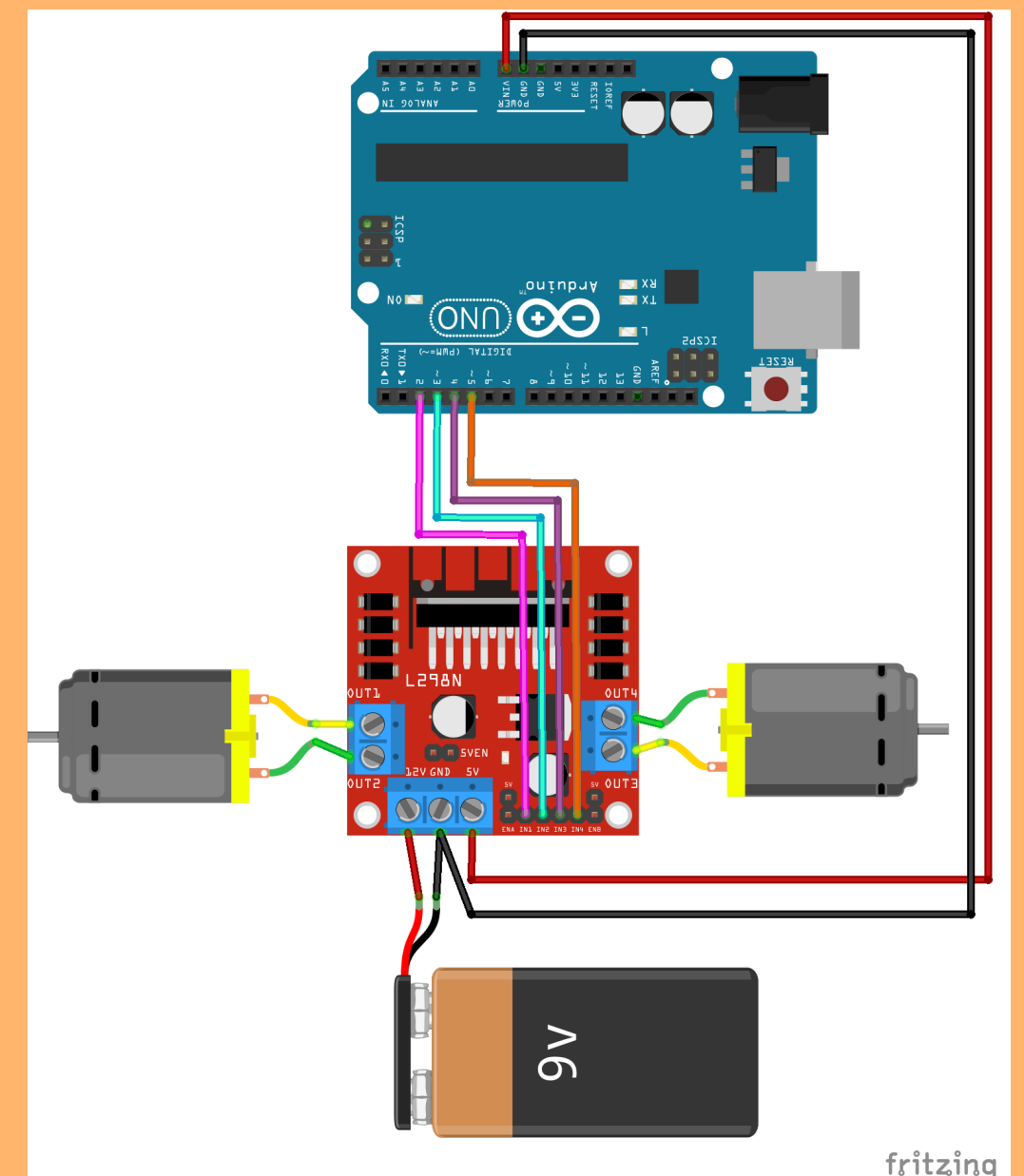
Led matrix is a dot matrix of large display, low resolution value and is useful for both industrial or commercial displays as well as for hobbyist human interface machines. In contains a 2-D diode matrix which have the cathode joined in rows and anode in columns. In this LED matrix each can be control individually by controlling the electricity through each pair of columns or rows diodes. These matrix are very popular matrix by means of displaying information it, allows the static and animated images and text.

The matrix is controlled by another arduino nano, and is programmed as such. Text messages are sent to the parent arduino, and the parent arduino dumps, the values to the child arduino. The child arduino, reads the value, and displays the characters/letters on the matrix screen. Refer to fig (e), for the schemactic.

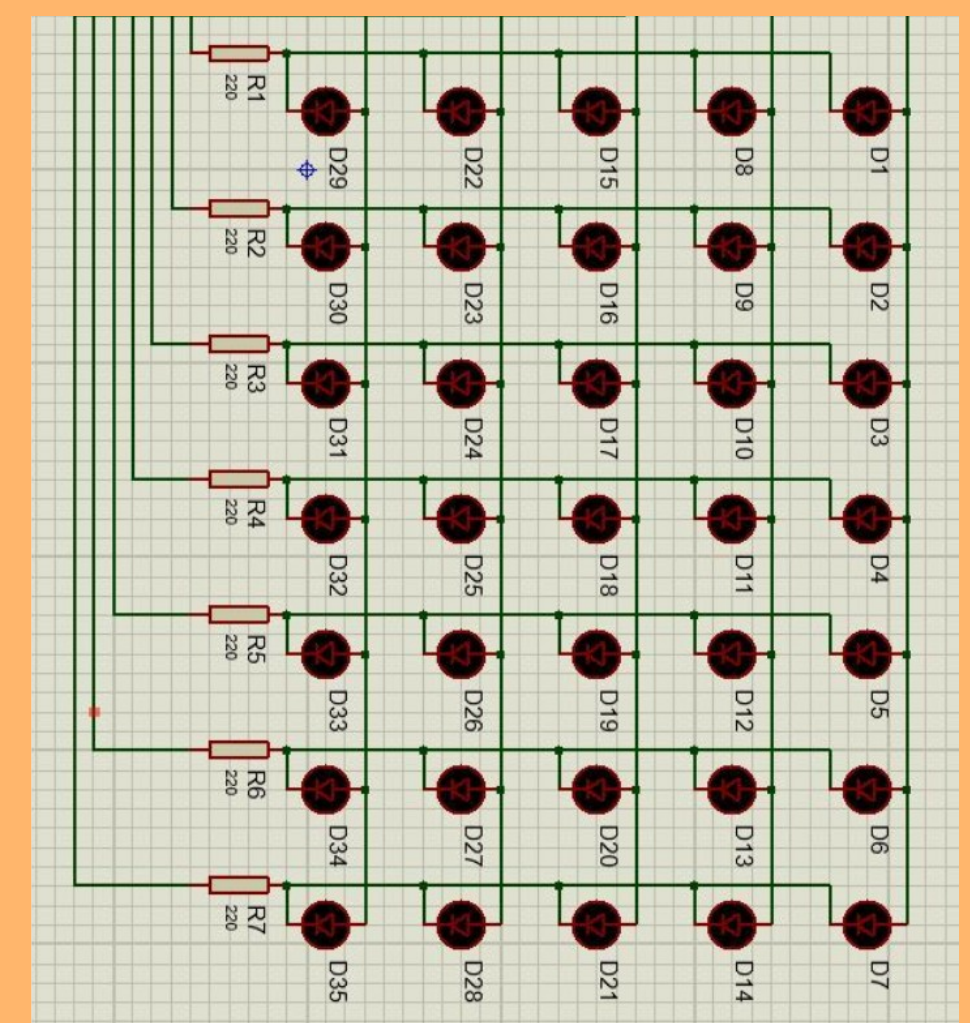
Project Members : Ritabrata Kabiraj and Ayan Mondal



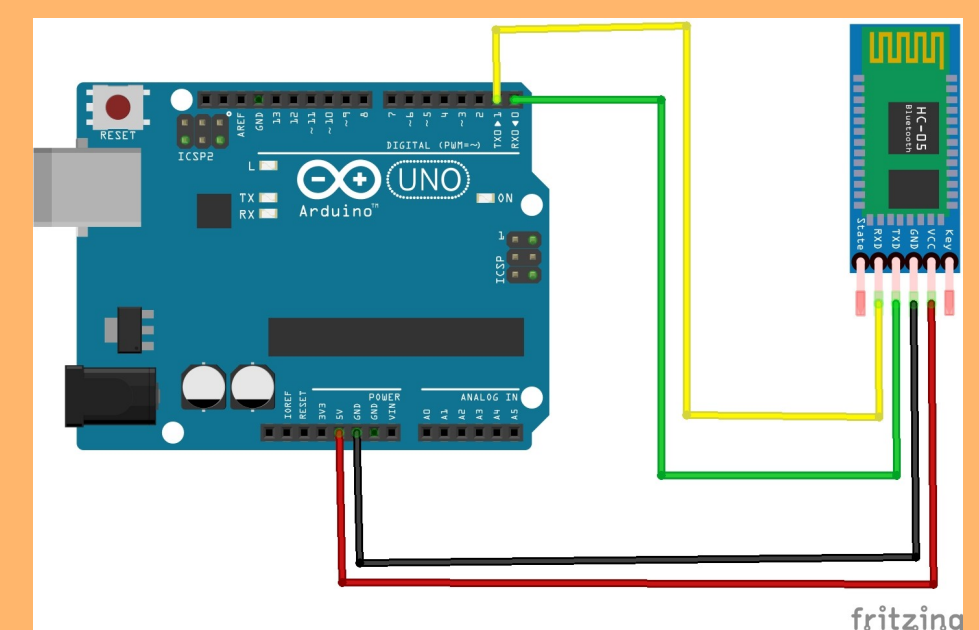
(Fig : Development Stage)



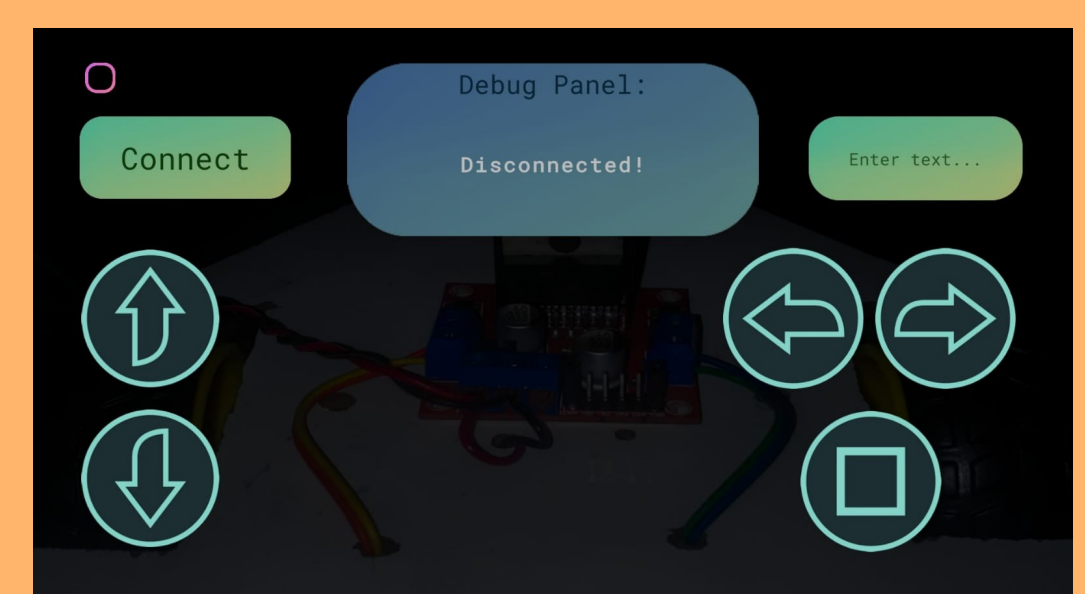
(Fig : a & b) (the red circuit is the L298N)



(Fig e : LED Matrix Schematic)



(Fig c : HC-05 Chip Schematic)



(Fig d : Snapshot of the Application)